



Twitter gets community notes

The feature will let you contextualise tweets to prevent misinformation on the app. <https://dgit.in/dec22-23>



Xiaomi launches a mini PC

It runs on a 12th generation Core i5-1240P processor and comes with Windows 11 pre-installed. <https://dgit.in/dec22-24>

solution by using the right and more efficient process nodes for the GCD and the MCD. On the 5nm node, the GCD manages to pack in a lot more firepower without a bump in the die size or power consumption. While the MCD uses a more efficient 6nm node that allows for a more cost-efficient approach.

BUT WHAT ABOUT THE INTERCONNECT?

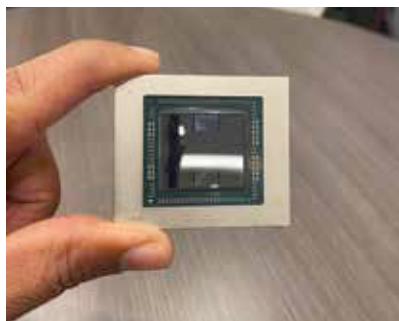
The biggest challenge that AMD faced when choosing the Chiplet route was to manage the bandwidth requirement along with tackling the latency issue. The interconnect was solved using a custom ultra-high bandwidth interface that AMD calls the Infinity fan out link interface that enables a staggering 5.3 TB/sec of throughput between the GCD and the MCDs.

TACKLING THE LATENCY FACTOR

The next critical problem related to the chiplet approach was to solve for the latency. AMD claims to have dealt with the latency factor by increasing the clock speeds. The GPU clock speeds were cranked up by 18 per cent to achieve an overall drop in latency by 10 per cent compared to the Navi 21 (Amd Radeon RX 6900 series).

A BIG JUMP IN THE GRAPHICS COMPUTE PERFORMANCE

The main graphics processor houses 48 WGP (workgroup processors) in the Navi 31 as compared to 40 WGP in Navi 21. And you still have two CUs (compute units) per WGP. So the total count of compute units have gone up from 80 in Navi 21 to 96 in Navi 31, a modest 20 per cent bump.



However, what has changed in a big way is the resources available for each CU. With the RDNA 3, each of the compute units gets twice the capability by having a pair each of ALUs and vector units.

BIG BOOST IN RAY TRACING PERFORMANCE

AMD had no choice but to bring some significant gains when it comes to Ray Tracing performance. With the RDNA 3, AMD has announced its second generation Ray Tracing Engine which the company claims improves ray tracing performance by up to 80

per cent. Quite a tall claim, we'd only be able to test it out once we have the cards in our test center. The approach to achieve higher RT performance is based on a smarter way of getting things done rather than using brute force. All in a bid to make things more efficient. AMD has optimized its second-generation ray-tracing engine to extract more performance out of each ray by using early subtree culling along with hardware support for DXR Ray Flags which significantly reduces the required instruction count per loop. Thereby achieving higher efficiency on RT workloads. **d**

We fielded some tough (and some not-so-tough) questions to the team behind the development of the new RDNA 3 architecture and here's what they had to say:

d How much of a cost-benefit does the chiplet design bring in terms of % over the conventional monolithic approach?

Sam Naffziger (Senior Vice President and Corporate Fellow, AMD):

I can't give you the specifics on that, our Navi 21 was 520mm square, and to go to 5nm with our Navi 31 design, it would be about that size or a little bigger. And the yields on advanced technology node is going to be an expensive buy and by splitting it, it's just a smarter way to design chips, especially as 6nm is very mature and highly optimized. The yield curves, there is an inflexion point, when you get to 400, 500 and 600mm square, the costs really take off.

d Will we see the chiplet design approach filtering down the mid-range cards?

Sam Naffziger (Senior Vice President and Corporate Fellow, AMD):

Like I said, the yield curve is non linear and there is a point where chiplets aren't of benefit. We are not talking about our future products but there is a point in our product line-up where a traditional design makes a better choice over modern designs.

d How closely is AMD working with the game development houses to optimize game performance for RDNA 3, given the chiplet approach, will it need additional optimization on the development and support side?

Laura Smith (CVP, Graphics MNC and Product Management, Radeon Technologies Group): We are broadly working with them to take advantage of RDNA 3, they don't really have to do anything additional from a support point of view, but we are working with them to tune software for RDNA 3 as a new architecture.